

Data Preparation Document: Cyclistic Bike-Share Demand Analysis

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Tool: Google BigQuery (SQL) → Tableau

Purpose

This document records the data-gathering and table-design phase of the Cyclistic project: how the required metrics and fields were identified from the Project Requirements and Strategy Documents, which datasets were combined, the transformation logic applied, and the resulting reporting table structure uploaded to Tableau for dashboard development.

Process Overview

- Reviewed the Project Requirements and Strategy Documents to identify the fields and grain needed to support the 7 required dashboard views (station demand, popular destinations, summer 2015 trend, year-over-year growth, station congestion, total trip volume, and peak usage/weather).
- Identified three data sources to combine: the Citi Bike trip data, a geographic boundary dataset to resolve station coordinates to zip code, and a weather dataset to capture daily conditions.
- Built a single SQL query in BigQuery to join, aggregate, and export one reporting table containing all fields required across the dashboard's chart set.
- Exported the resulting table for upload into Tableau to build the final dashboard.

Data Sources

Source	Table	Role
Primary — trip data	bigquery-public-data.new_york_citibike.citibike_trips	Trip-level records: start/end time, station coordinates, bike ID, user type, trip duration
Secondary — station geography	bigquery-public-data.geo_us_boundaries.zip_codes	Resolves station latitude/longitude to a zip code via a spatial (ST_WITHIN) join
Secondary — zip lookup	extreme-splicer-475811-t7.Cyclistic.zip_codes	Custom-uploaded table mapping zip code → borough → neighborhood, for human-readable geography
Supplementary — weather	bigquery-public-data.noaa_gsod.gsod20*	Daily mean temperature, wind speed, and total precipitation from the Central Park station (WBAN 94728)

Transformation Logic

- Spatial join: each trip's start and end coordinates are matched to a zip code polygon using ST_WITHIN(ST_GEOPOINT(...), zip_code_geom) — run once for the start station and once for the end station.
- Geography lookup: each resolved zip code is joined against the custom zip_codes table to attach borough and neighborhood names.

- Weather join: each trip's start date is matched to that day's weather record from the NOAA GSOD dataset.
- Date shift: 5 years are added to all trip dates (DATE_ADD ... INTERVAL 5 YEAR) so the historical 2014–2015 dataset displays as recent activity in the fictional dashboard scenario.
- Trip duration is converted from seconds to minutes and rounded to the nearest 10-minute bucket, to reduce row count and support duration-based aggregation.
- Data is filtered to the Central Park weather station and to trips from 2014–2015, then grouped by all non-aggregate fields, with COUNT(bikeid) producing a trip_count roll-up rather than exporting every individual trip row.

SQL Query

```
SELECT
  TRI.usertype,
  ZIPSTART.zip_code AS zip_code_start,
  ZIPSTARTNAME.borough AS borough_start,
  ZIPSTARTNAME.neighborhood AS neighborhood_start,
  ZIPEND.zip_code AS zip_code_end,
  ZIPENDNAME.borough AS borough_end,
  ZIPENDNAME.neighborhood AS neighborhood_end,
  DATE_ADD(DATE(TRI.starttime), INTERVAL 5 YEAR) AS start_day,
  DATE_ADD(DATE(TRI.stoptime), INTERVAL 5 YEAR) AS stop_day,
  WEA.temp AS day_mean_temperature,
  WEA.wdsp AS day_mean_wind_speed,
  WEA.prcp AS day_total_precipitation,
  ROUND(CAST(TRI.tripduration / 60 AS INT64), -1) AS trip_minutes,
  COUNT(TRI.bikeid) AS trip_count
FROM
  `bigquery-public-data.new_york_citibike.citibike_trips` AS TRI
INNER JOIN
  `bigquery-public-data.geo_us_boundaries.zip_codes` ZIPSTART
  ON ST_WITHIN(
    ST_GEOPOINT(TRI.start_station_longitude, TRI.start_station_latitude),
    ZIPSTART.zip_code_geom)
INNER JOIN
  `bigquery-public-data.geo_us_boundaries.zip_codes` ZIPEND
  ON ST_WITHIN(
    ST_GEOPOINT(TRI.end_station_longitude, TRI.end_station_latitude),
    ZIPEND.zip_code_geom)
INNER JOIN
  `bigquery-public-data.noaa_gsod.gsod20*` AS WEA
  ON PARSE_DATE("%Y%m%d", CONCAT(WEA.year, WEA.mo, WEA.da)) = DATE(TRI.starttime)
INNER JOIN
  `extreme-splicer-475811-t7.Cyclistic.zip_codes` AS ZIPSTARTNAME
  ON ZIPSTART.zip_code = CAST(ZIPSTARTNAME.zip AS STRING)
INNER JOIN
  `extreme-splicer-475811-t7.Cyclistic.zip_codes` AS ZIPENDNAME
  ON ZIPEND.zip_code = CAST(ZIPENDNAME.zip AS STRING)
WHERE
  WEA.wban = '94728'
  AND EXTRACT(YEAR FROM DATE(TRI.starttime)) BETWEEN 2014 AND 2015
GROUP BY
  1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13
```

Output: Reporting Table Schema

Field	Type	Description
usertype	STRING	Subscriber or Customer (one-time rider)
zip_code_start / zip_code_end	STRING	Zip code of the trip's start / end station

borough_start / borough_end	STRING	Borough of the start / end zip code
neighborhood_start / neighborhood_end	STRING	Neighborhood of the start / end zip code
start_day / stop_day	DATE	Trip start / end date (shifted +5 years for the fictional dashboard)
day_mean_temperature	FLOAT	Mean temperature on the trip's start date
day_mean_wind_speed	FLOAT	Mean wind speed on the trip's start date
day_total_precipitation	FLOAT	Total precipitation on the trip's start date
trip_minutes	INTEGER	Trip duration in minutes, rounded to the nearest 10
trip_count	INTEGER	Number of trips matching this combination of fields

Mapping to Dashboard Requirements

Dashboard Chart	Supported By
Station demand by starting location	zip_code_start, borough_start, neighborhood_start, trip_count
Popular destinations by trip minutes (peak months)	zip_code_end / neighborhood_end, trip_minutes, start_day
Summer 2015 trend	start_day filtered to Jun-Aug 2015 (pre-shift), trip_count
Year-over-year % growth	start_day (year), trip_count
Station congestion (net flow)	zip_code_start vs. zip_code_end, start_day, trip_count
Total trips across all locations	SUM(trip_count)
Peak usage by time/season/weather	start_day, day_mean_temperature, day_total_precipitation, trip_count
Subscriber vs. one-time customer breakdown	usertype, trip_count

Next Step

The reporting table produced by this query has been exported and is ready to upload into Tableau, where it will power the dashboard defined in the Strategy Document.